Alvin (Haoyun) Yang

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Research Interests

Theoretical High-Energy Physics, Fundamental Quantum Field Theory (QFT), Particle Phenomenology, Gravity and Quantum Information.

Education

University of Illinois Urbana-Champaign (UIUC) Urbana, IL
Bachelor of Science in Physics Expected: May 2027

- **Technical GPA:** 3.91 / 4.00
- **Honors/Awards:** [Grainger College of Engineering Achievement Scholarship, FA25, FA26 Dean's List, James Scholar]

Research Experience

Project: Minimax Geometry and Morse-Type Topology of Competing RT Surfaces in AdS_3 UIUC Department of Physics
Advisor: Prof. Thomas Faulkner [Fall 2026] – Present
PHYS 497: Independent Study (3 credits)

- Studied the variational formulation of RT/HRT surfaces and Wall's Maximin construction, with related readings on FLM corrections and entanglement-wedge reconstruction.
- Constructed a finite-dimensional broken-geodesic configuration space for competing connected and disconnected RT pairings on a time-symmetric AdS_3 slice, gluing two four-dimensional branches along their common two-dimensional locus of four-arm configurations.
- Derived an exact restricted minimax length between the two RT configurations,

$$\mathcal{L}_{\text{minimax}} = d_{\mathbb{H}^2}(p_1, p_3) + d_{\mathbb{H}^2}(p_2, p_4),$$

  using the hyperbolic triangle inequality and geodesic convexity.
- Characterized the topology of length sublevel sets, identifying index-0-type component births and an index-1-type merger at the minimax level; computed the associated relative homology, with

$$H_1(\mathcal{X}^{h_+}, \mathcal{X}^{h_-}; \mathbb{Z}_2) \cong \mathbb{Z}_2.$$

- Investigating whether minimax data can capture geometric information beyond competing RT minimal lengths, including the response to localized metric deformations and their possible physical realization.

Project: Scattering Amplitudes and Viscosity Cross Sections in Light-Mediator SIDM UIUC Department of Physics
Advisor: Prof. Tanner Trickle [Spring 2026] – Present
Undergraduate Research Assistant

- Investigated the regime of validity of the Born-approximation formula in Eq. (B2) of Knapen, Lin, and Zurek (arXiv:1709.07882). Computed spin-averaged tree-level squared amplitudes for seven leading-order scattering processes using Mathematica/FeynCalc, resolving fermion-exchange signs, neutrino helicity averaging, and global substitution contamination in `SetMandelstam`.
- Performed systematic asymptotic analysis in the dimensionless parameter $R = m_\chi v / m_{A'}$, identifying an unphysical R^2 divergence of the conventional momentum-transfer cross section in identical-particle scattering, originating from the unsuppressed backward u-channel pole.
- Derived the viscosity angular weighting $\sin^2\theta$ from forward-backward symmetry and the traceless symmetric second-order velocity moment of the Boltzmann collision integral, connecting the resulting weighting to the quadrupolar ($\ell = 2$) angular structure.
- Demonstrated that the viscosity cross section removes the wrong scale of light-mediator behaviors and reproduces the leading asymptotic scaling of Eq. (B2). Analytically traced the normalization factor of 8 to the quadrupolar tensor-moment integration and identical-particle phase-space symmetry factor.

Future Scientist Exchange Program (FuSEP 25) Participants University of Science and Technology of China (USTC) Department of Modern Physics

Advisor: Prof. Shi Pu [July 2025]
Project: An algebraic satisfied and gauge invariant decomposition of Angular Momentum

- Investigated decompositions of angular momentum(Canonical, Belinfante's, and Chen et al.'s formulations).
- Reproduced the idea and derivation of Chen et al's decomposition in a QED system.
- Presented a comprehensive technical poster at the final FuSEP 25 conference.

Mathematical & Theoretical Background

Academic Baselines & Completed Courses

- **Abstract Algebra** **Grade: A+**
  Core Mastery: Fundamental theorem of arithmetic, congruences. Permutations. Groups and subgroups, homomorphisms. Group actions with applications. Polynomials. Rings, subrings, and ideals. Integral domains and fields. Roots of polynomials. Maximal ideals, construction of fields.
- **Complex Analysis** **Grade: A+**
  Core Mastery: Covers the standard topics and gives an introduction to integration by residues, the argument principle, conformal maps, and potential fields.
- **Undergraduate Core Foundations:** Classical Mechanics (A+), Quantum Mechanics (A+), Electrodynamics (A+), Statistical Mechanics (A+).

Concurrent Advanced Enrollment (Fall 2026)

- **PHYS 515: General Relativity I (Graduate Level)**
  Focus: Systematic introduction to Einstein's theory, with emphasis on modern coordinate-free methods of computation. Review of special relativity, modern differential geometry, foundations of general relativity, laws of physics in the presence of a gravitational field, linearized theory, and experimental tests of gravitation theories.
- **MATH 428: Introductory Morse Theory (Honors)**
  Focus: Basic of Morse Theory, Influential ideas of Smale, Witten and Floer, as much geodesics of Riemannian metrics and culminate in the celebrated Periodicity Theorem of Bott as possible.
- **PHYS 508: Mathematical Physics** *(Auditing)*
  Focus: Core techniques of mathematical physics widely used in the physical sciences. Calculus of variations and its applications; partial differential equations of mathematical physics (including classification and boundary conditions); separation of variables, series solutions of ordinary differential equations and Sturm-Liouville eigenproblems; Legendre polynomials, spherical harmonics, Bessel functions and their applications; normal mode eigenproblems (including the wave and diffusion equations); inhomogeneous ordinary differential equations (including variation of parameters); inhomogeneous partial differential equations and Green functions; potential theory; integral equations (including Fredholm theory).
- **PHYS 580: Quantum Mechanics (Graduate Level)** *(Auditing)*
  Focus: Operators, state vectors, and the formal structure of quantum theory; operator treatments of simple systems; angular momentum and vector addition coefficients; stationary state perturbation theory; introduction to scattering theory for particles without spin, partial wave analysis, and Born approximation; examples taken from atomic, nuclear, and elementary particle physics.

[1] Last updated: Nov 2026